Music Booking Platform - ERD Description and Sample Data

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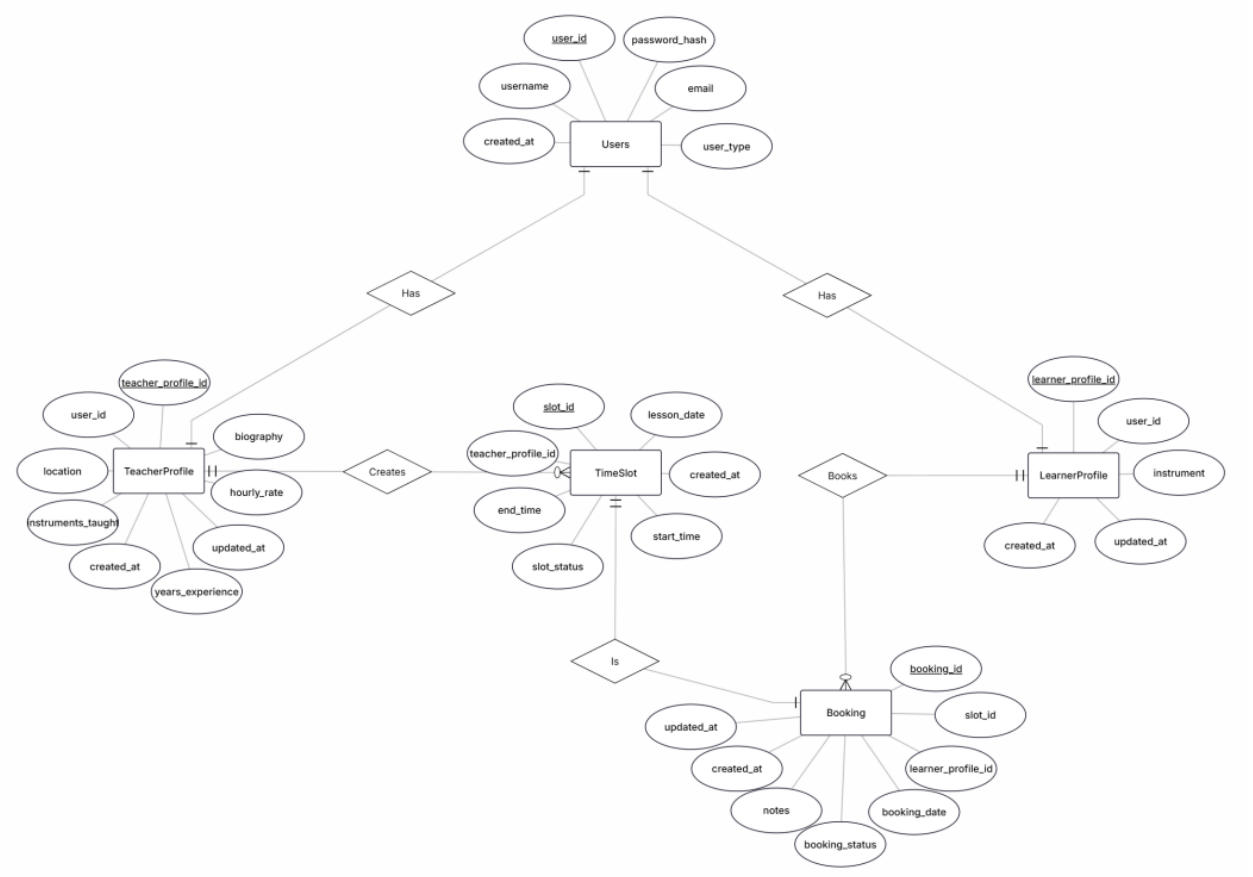
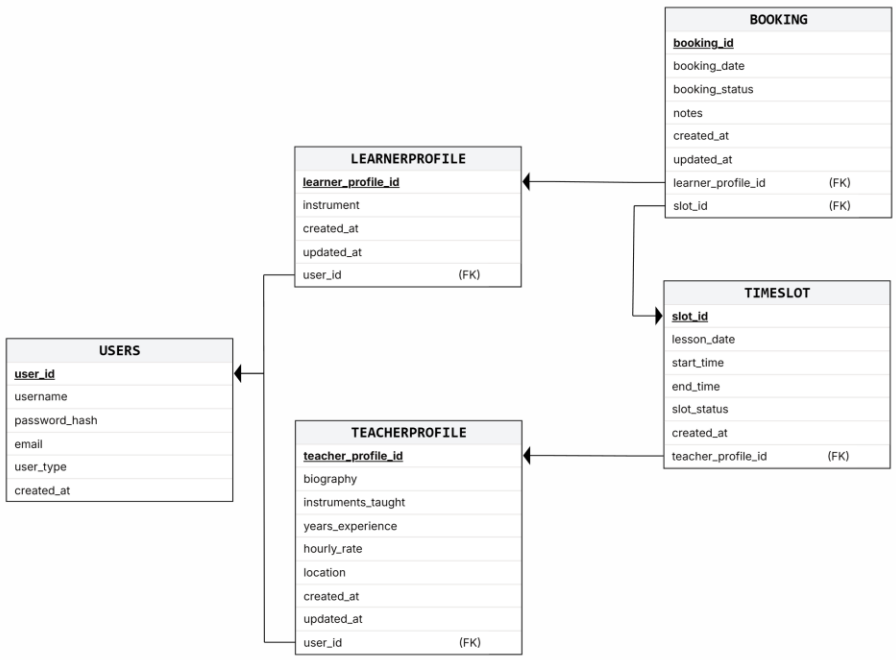
1. Project Vision

Our platform aims to simplify the process of finding and booking private music lessons for both learners and teachers. Students can easily search for qualified teachers based on instrument, schedule, location, and price. Teachers can create detailed profiles to showcase their skills, set their availability, and manage bookings seamlessly through a dedicated interface. The goal is to build a user-friendly system that fosters music education for learners of all ages while creating flexible teaching opportunities for freelance musicians.

2. Entity Relationship Diagram (ERD)



3. Relational Database Schema



4. Verbal Description of Each Entity and Relationship

Entities

USERS

Description: Represents all registered users in the system, including both teachers and learners. Stores login information and user type.

Key Attributes: user_id (Primary Key), username, password_hash, email, user_type (‘teacher’ or ‘student’), created_at

TEACHERPROFILE

Description: Stores detailed information about teachers, including teaching skills, instruments taught, experience, hourly rate, and location. Each teacher corresponds to exactly one USERS account.

Key Attributes: teacher_profile_id (Primary Key), user_id (Foreign Key), biography, instruments_taught, years_experience, hourly_rate, location, created_at, updated_at

LEARNERPROFILE

Description: Stores detailed information about learners (students or adults), including the instrument they want to learn. Each learner corresponds to exactly one USERS account.

Key Attributes: learner_profile_id (Primary Key), user_id (Foreign Key), instrument, created_at, updated_at

TIMESLOT

Description: Represents the time slots published by teachers for lessons. Each time slot is linked to a specific teacher.

Key Attributes: slot_id (Primary Key), teacher_profile_id (Foreign Key), lesson_date, start_time, end_time, slot_status (available/booked), created_at

BOOKING

Description: Represents lesson bookings by learners for specific time slots. Linked to both the learner and the selected time slot.

Key Attributes: booking_id (Primary Key), slot_id (Foreign Key), learner_profile_id (Foreign Key), booking_date, booking_status, notes, created_at, updated_at

Relationships

USERS → TEACHERPROFILE

Type: 1:1

Description: Each teacher user has exactly one teacher profile.

USERS → LEARNERPROFILE

Type: 1:1

Description: Each learner user has exactly one learner profile.

TEACHERPROFILE → TIMESLOT

Type: 1:N

Description: One teacher can create multiple lesson time slots.

LEARNERPROFILE → BOOKING

Type: 1:N

Description: One learner can book multiple lessons.

TIMESLOT → BOOKING

Type: 1:1

Description: A booked time slot corresponds to exactly one booking record. Once a slot is booked, it becomes unavailable for others. This ensures no double-booking occurs for the same time slot.

5. Sample Data for Each Table

USERS

user_id	username	password_hash	email	user_type	created_at
1	alice	hash1	alice@mail.com	teacher	2026-02-20
2	bob	hash2	bob@mail.com	student	2026-02-20
3	carol	hash3	carol@mail.com	teacher	2026-02-21
4	david	hash4	david@mail.com	student	2026-02-21

TEACHERPROFILE

teacher_profile_id	user_id	biography	instruments_taught	years_experience	hourly_rate	location	created_at	updated_at
1	1	Northwestern Newcastle University Professor	Piano	10	50	Helsinki	2026-02-20	2026-02-20
2	3	12 years of experience teaching guitar	Guitar	5	40	Espoo	2026-02-21	2026-02-21

LEARNERPROFILE

learner_profile_id	user_id	instrument	created_at	updated_at
1	2	Piano	2026-02-20	2026-02-20
2	4	Guitar	2026-02-21	2026-02-21

TIMESLOT

slot_id	teacher_profile_id	lesson_date	start_time	end_time	slot_status	created_at
1	1	2026-02-25	09:00	10:00	available	2026-02-20
2	1	2026-02-25	10:00	11:00	available	2026-02-20
3	2	2026-02-26	14:00	15:00	available	2026-02-21

BOOKING

booking_id	slot_id	learner_profile_id	booking_date	booking_status	notes	created_at	updated_at
1	1	1	2026-02-20	confirmed	Null	2026-02-20	2026-02-20
2	3	2	2026-02-21	confirmed	First lesson	2026-02-21	2026-02-21